PRD: Kidzee Polichalur MVP 2 — Visibility & Auto-Publishing

Introduction

MVP 1 delivered a centralized activity platform with admin CRUD and share links. MVP 2 closes the remaining customer requirements: **"When we update them here it would be great if they get uploaded in Google, YouTube, Insta and FB or similar platforms for visibility."**

MVP 2 uses pragmatic defaults — build what we can without waiting for Imtiaz's Batch 2 questionnaire responses. Every feature is designed to work in "manual-first" mode and upgrade to "auto" mode once API credentials are provided.

Customer: Imtiaz (Kidzee Polichalur franchise owner) **Source:** WhatsApp conversation, 28 March 2026
MVP 1 Status: Live at https://kidzee-polichalur.vercel.app

Goals

- Make activities discoverable on Google search ("Kidzee Polichalur" queries)
- Enable social media posting to Instagram and Facebook (queue → approve → publish pipeline)
- Support YouTube video embedding and deep linking
- Add image upload capability for activity photos
- Migrate data storage from fragile JSON files to Supabase
- Track all costs, effort, and API usage for client billing transparency

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Cost & Effort Tracker

Estimated Effort

Story	Effort (hrs)	Complexity	Dependencies
US-001: SEO meta tags + Open Graph	1.5	Low	None
US-002: Sitemap + robots.txt	1	Low	None
US-003: JSON-LD structured data	1.5	Low	US-001
US-004: Supabase migration	3	Medium	Supabase project
US-005: Image upload	3	Medium	US-004 (Supabase Storage)
US-006: Meta API integration	4	High	Meta Business account
US-007: Social publish pipeline	2.5	Medium	US-006
US-008: YouTube embed + deep link	1.5	Low	None
US-009: Activity detail page	2	Medium	US-005
US-010: MVP 2 documentation	1.5	Low	All above
Total	**22 hrs**		

API & Service Costs

Service	Free Tier	Monthly Cost	Notes
Vercel (hosting)	100GB bandwidth	Rs. 0	Current plan sufficient
Supabase (database + storage)	500MB DB, 1GB storage	Rs. 0	Free tier covers MVP 2
Meta Graph API	Unlimited	Rs. 0	Free, requires app review
YouTube Data API	10,000 units/day	Rs. 0	Free, requires Google OAuth
Google Search Console	Unlimited	Rs. 0	Free
Custom domain (.in)	N/A	Rs. 500-1500/year	Optional, pending Imtiaz response
Total monthly		**Rs. 0**	All free tier

One-Time Setup Costs

Item	Cost	Notes
Meta App Review	Rs. 0	Free but takes 5-10 business days
Google Cloud project (YouTube API)	Rs. 0	Free
Supabase project	Rs. 0	Free tier
Domain registration (if needed)	Rs. 500-1500	Pending Imtiaz
Total one-time	**Rs. 0-1500**	

User Stories

US-001: SEO Meta Tags + Open Graph

Description: As a parent searching Google for "Kidzee Polichalur," I want the platform to appear in search results with a rich preview so I can find and share it easily.

Acceptance Criteria:

- [ ] Every page has unique `<title>` and `<meta name="description">`
- [ ] Open Graph tags on all pages (`og:title` , `og:description` , `og:image` , `og:url` , `og:type`)
- [ ] Twitter Card meta tags (`twitter:card` , `twitter:title` , `twitter:description` , `twitter:image`)
- [ ] Homepage title: "Kidzee Polichalur — Preschool Activities & Events"
- [ ] Activities page title: "Activities — Kidzee Polichalur"
- [ ] Year page title: "2024 Activities — Kidzee Polichalur"
- [ ] OG image defaults to Kidzee logo (can be overridden per activity)

- [ ] `npm run build` passes
- [ ] Validate with https://metatags.io or Facebook Sharing Debugger

Files: `src/app/layout.tsx`, `src/app/page.tsx`, `src/app/activities/page.tsx`, `src/app/activities/[year]/page.tsx` **Effort:** 1.5 hrs

US-002: Sitemap + Robots.txt

Description: As Google's crawler, I need a sitemap.xml and robots.txt so I can discover and index all activity pages.

Acceptance Criteria:

- [ ] `sitemap.xml` generated dynamically at `/sitemap.xml` via Next.js `sitemap.ts`
- [ ] Includes: homepage, `/activities`, each `/activities/{year}` page
- [ ] `robots.txt` at `/robots.txt` via Next.js `robots.ts`
- [ ] Robots allows all crawlers, points to sitemap URL
- [ ] Admin pages excluded from sitemap (not `/admin`, `/admin/login`)
- [ ] API routes excluded from sitemap
- [ ] `npm run build` passes

Files: `src/app/sitemap.ts`, `src/app/robots.ts` **Effort:** 1 hr

US-003: JSON-LD Structured Data

Description: As a search engine, I need structured data (Schema.org) to understand the content and display rich results.

Acceptance Criteria:

- [ ] Homepage: `Organization` schema with name, logo, address, URL
- [ ] Homepage: `EducationalOrganization` type with `preschool` descriptor
- [ ] Each activity: `Event` schema with name, date, description, location
- [ ] Breadcrumb schema on activity pages
- [ ] Validate with Google Rich Results Test (https://search.google.com/test/rich-results)
- [ ] `npm run build` passes

Files: `src/components/structured-data.tsx`, `src/app/page.tsx`, `src/app/activities/[year]/page.tsx` **Effort:** 1.5 hrs

US-004: Supabase Migration

Description: As the platform, I need persistent database storage so data survives Vercel serverless cold starts and supports image uploads.

Acceptance Criteria:

- [ ] Create Supabase project for Kidzee Polichalur (or use existing zoo-crm Supabase)
- [ ] Create `activities` table matching current Activity schema
- [ ] Create `social_posts` table matching current SocialPost schema
- [ ] Migrate `src/lib/data.ts` from JSON file I/O to Supabase client

- [ ] Migrate `src/lib/social-posts.ts` from JSON to Supabase
- [ ] Seed existing 10 demo activities into Supabase
- [ ] All existing CRUD operations work unchanged (GET, POST, DELETE)
- [ ] Environment variables: `SUPABASE_URL`, `SUPABASE_ANON_KEY`
- [ ] RLS policies: public read, authenticated write
- [ ] `npm run build` passes
- [ ] Verify in browser: activities load, admin can add/delete

Files: `src/lib/data.ts`, `src/lib/social-posts.ts`, `src/lib/supabase.ts` (new), migration SQL **Effort:** 3 hrs **Dependency:** Supabase project setup

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US-005: Image Upload for Activities

Description: As an admin, I want to upload photos when creating an activity so parents can see what happened.

Acceptance Criteria:

- [ ] Image upload field in admin "Add Activity" form (multi-file, max 10 images)
- [ ] Accept JPEG, PNG, WebP (max 5MB each)
- [ ] Upload to Supabase Storage bucket `activity-images`
- [ ] Store image URLs in activity.images array
- [ ] Show image thumbnails in activity cards (first image as hero, rest as count badge)
- [ ] Image gallery/lightbox on activity detail page (or expand on card)
- [ ] Images included in Open Graph tags (`og:image` = first activity image)
- [ ] Admin can remove uploaded images
- [ ] `npm run build` passes
- [ ] Verify in browser: upload images, see them on activity cards

Files: `src/app/admin/page.tsx`, `src/app/api/activities/route.ts`, `src/components/activity-card.tsx`, `src/components/image-upload.tsx` (new) **Effort:** 3 hrs **Dependency:** US-004 (Supabase Storage)

US-006: Meta Graph API Integration

Description: As the platform, I need to connect to Meta's Graph API so activities can be published to Facebook Pages and Instagram Business accounts.

Note: This feature works in "manual-first" mode until Imtiaz provides Meta Business credentials. The UI is built, but actual publishing requires API credentials configured in environment variables.

Acceptance Criteria:

- [ ] Create `/api/social/publish` endpoint that posts to Meta Graph API
- [ ] Support Facebook Page posts: text + images via `/{page-id}/photos` or `/{page-id}/feed`
- [ ] Support Instagram Business posts: image + caption via `/{ig-user-id}/media` + `/{ig-user-id}/media_publish`
- [ ] Handle API errors gracefully (token expired, rate limit, permissions)
- [ ] Update social post status: "queued" → "publishing" → "published" / "failed"

- [ ] Store Meta post IDs in social_posts record for reference
- [ ] Works without credentials (graceful fallback: "Meta API not configured" message)
- [ ] Environment variables: `META_PAGE_ACCESS_TOKEN`, `META_PAGE_ID`, `META_IG_USER_ID`
- [ ] `npm run build` passes

Files: `src/app/api/social/publish/route.ts` (new), `src/lib/meta-api.ts` (new), `src/lib/social-posts.ts` **Effort:** 4 hrs **Dependency:** Meta Business account (from Imtiaz)

US-007: Social Publish Pipeline UI

Description: As an admin, I want to review, approve, and publish social media posts from the dashboard instead of just queuing them.

Acceptance Criteria:

- [ ] "Social Posts" section in admin dashboard showing queued/published/failed posts
- [ ] Each queued post shows: caption preview, platform, created date, activity title
- [ ] "Publish Now" button on each queued post (calls `/api/social/publish`)
- [ ] "Edit Caption" ability before publishing
- [ ] Status badges: queued (yellow), published (green), failed (red)
- [ ] Published posts show platform post URL (link to Facebook/Instagram post)
- [ ] Failed posts show error message + "Retry" button
- [ ] If Meta API not configured: show "Copy Caption" button instead of "Publish"
- [ ] `npm run build` passes
- [ ] Verify in browser: queue a post, see it in list, publish/copy

Files: `src/app/admin/page.tsx`, `src/components/social-post-list.tsx` (new) **Effort:** 2.5 hrs **Dependency:** US-006

US-008: YouTube Embed + Deep Link

Description: As a parent, I want to watch activity videos directly on the platform and as an admin, I want to easily link YouTube videos.

Acceptance Criteria:

- [ ] YouTube URL field in admin form auto-detects video ID from various URL formats (youtube.com/watch, youtu.be, shorts)
- [ ] Activity cards show video thumbnail from YouTube API (`https://img.youtube.com/vi/{id}/hqdefault.jpg`)
- [ ] Play button overlay on video thumbnail
- [ ] Click opens embedded YouTube player (iframe) inline or in modal
- [ ] Video embed is responsive (16:9 aspect ratio)
- [ ] OG meta tags include `og:video` for activities with YouTube links
- [ ] `npm run build` passes
- [ ] Verify in browser: add YouTube URL, see thumbnail, click to play

Files: `src/components/activity-card.tsx`, `src/components/youtube-embed.tsx` (new), `src/lib/youtube.ts` (new) **Effort:** 1.5 hrs

US-009: Activity Detail Page

Description: As a parent sharing an activity link, I want it to open a full dedicated page (not just an anchor on a list) with all photos, video, and details.

Acceptance Criteria:

- [ ] New route: `/activities/{year}/{id}` — full-page activity view
- [ ] Shows: title, date, category, full description, all images (gallery), video embed, tags
- [ ] Share buttons (same as card share panel)
- [ ] Social auto-post button
- [ ] Proper SEO meta tags (unique title, description, OG image from first activity image)
- [ ] JSON-LD Event schema for the specific activity
- [ ] Back button → returns to activities list
- [ ] Activity cards link to this detail page
- [ ] `npm run build` passes
- [ ] Verify in browser: click activity → see detail page → share → navigate back

Files: `src/app/activities/[year]/[id]/page.tsx` (new), `src/components/activity-gallery.tsx` (new) **Effort:** 2 hrs **Dependency:** US-005 (images), US-008 (video)

US-010: MVP 2 Documentation

Description: As a stakeholder, I want updated documentation reflecting all MVP 2 features.

Acceptance Criteria:

- [ ] Update `docs/stakeholder-docs/STAKEHOLDER-REVIEW.md` with MVP 2 features
- [ ] New screenshots for: image upload, social publish pipeline, activity detail page, YouTube embed
- [ ] Update API reference with new endpoints
- [ ] Update roadmap (MVP 2 items → "Done", add MVP 3 items)
- [ ] Regenerate PDF
- [ ] Update cost tracker with actual hours spent
- [ ] `npm run build` passes (all apps in monorepo)

Files: `docs/stakeholder-docs/STAKEHOLDER-REVIEW.md`, `docs/stakeholder-docs/screenshots/` **Effort:** 1.5 hrs **Dependency:** All above stories complete

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Functional Requirements

- FR-1: Every public page must have unique SEO meta tags (title, description, OG, Twitter Card)
- FR-2: Dynamic `sitemap.xml` must include all public pages and exclude admin/API routes
- FR-3: JSON-LD structured data on homepage (Organization) and activities (Event)
- FR-4: All data must be stored in Supabase (not JSON files)
- FR-5: Admin can upload up to 10 images per activity (max 5MB each, JPEG/PNG/WebP)
- FR-6: Images stored in Supabase Storage with public URLs
- FR-7: Platform can publish to Facebook Pages via Meta Graph API (when configured)
- FR-8: Platform can publish to Instagram Business via Meta Graph API (when configured)
- FR-9: Social posts follow pipeline: queued → approved → published/failed

- FR-10: YouTube videos embedded inline with responsive player
- FR-11: Each activity has a dedicated detail page at `/activities/{year}/{id}`
- FR-12: All features degrade gracefully when API credentials are not configured

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Non-Goals (Out of Scope for MVP 2)

- Direct YouTube video upload (admin pastes link; direct upload is MVP 3)
- Google Business Profile integration (MVP 3)
- Parent login/accounts (MVP 3)
- WhatsApp notifications to parents (MVP 3)
- Email newsletters (MVP 3)
- Analytics/engagement tracking (MVP 3)
- Multi-language support (MVP 3)
- PWA/offline capability (MVP 3)
- Custom domain setup (pending Imtiaz's response on budget)
- Automated scheduling (post at specific time) — MVP 2 is "publish now"
- Image watermarking with school logo (MVP 3)
- Activity edit/update (only add/delete in MVP 2; edit is MVP 3)

Technical Considerations

Supabase Setup

- Use existing zoo-crm Supabase project or create dedicated Kidzee project
- Tables: `activities`, `social_posts`
- Storage bucket: `activity-images` (public read, authenticated write)
- RLS: public SELECT on both tables, authenticated INSERT/DELETE

Meta Graph API

- Requires Meta Business account + Facebook Page + Instagram Business
- Page Access Token (long-lived, 60-day) stored in env vars
- Instagram publishing requires image URL (not upload) — use Supabase Storage public URLs
- Rate limits: 200 posts/hour (more than sufficient)
- App review required for `pages_manage_posts` and `instagram_content_publish` permissions

YouTube

- No API needed for embedding — just parse video ID from URL
- Thumbnail via `https://img.youtube.com/vi/{VIDEO_ID}/hqdefault.jpg`
- Embed via `https://www.youtube.com/embed/{VIDEO_ID}`

SEO

- Next.js 16 `generateMetadata()` for dynamic meta tags
- `sitemap.ts` and `robots.ts` in app directory (native Next.js support)
- JSON-LD via `<script type="application/ld+json">` in page components

Success Metrics

Metric	Target	How to Measure

Google indexing	All public pages indexed within 2 weeks of Search Console submission	Google Search Console → Coverage
Social post publishing	Admin can publish to FB/Insta in < 3 clicks from dashboard	Manual testing
Image upload	Admin can add photos in < 30 seconds per activity	Manual testing
Page load	All pages load in < 2 seconds on 4G mobile	Lighthouse audit
Uptime	No data loss on Vercel restarts	Verify activities persist after redeployment

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Implementation Order

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Phase 1 (No external deps — build immediately):
  US-001: SEO meta tags ──┐
  US-002: Sitemap/robots ──┼── Can ship independently
  US-003: JSON-LD ─────────┘
  US-008: YouTube embed ────── No API needed

Phase 2 (Supabase setup required):
  US-004: Supabase migration ──┐
  US-005: Image upload ────────┘── Sequential

Phase 3 (Meta credentials required — build UI, mock API):
  US-006: Meta API integration ──┐
  US-007: Social publish pipeline ┘── Sequential

Phase 4 (Depends on all above):
  US-009: Activity detail page
  US-010: Documentation update
```

Open Questions (Resolved by Batch 2 Questionnaire)

#	Question	Default (if no response)	Impact
1	Does Imtiaz have a Meta Business account?	Build UI anyway, add credentials later	US-006 scope
2	Does Imtiaz have a YouTube channel?	Embed-only (no upload)	US-008 scope
3	Does Imtiaz want a custom domain?	Keep vercel.app subdomain	SEO impact
4	Monthly budget?	Rs. 0 (free tier only)	Service selection
5	Auto-post or approve-first?	Approve-first (safer)	US-007 flow
6	Who reviews content?	Admin only	Access control

Version History

Version	Date	Changes
1.0.0	2026-03-29	Initial PRD — 10 user stories, 22 hrs estimated

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